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Profile Summary:

Devesh Saini is a final-year B.Tech Computer Science and Engineering student at Chandigarh Group of Colleges, Landran (2022–2026), with a CGPA of 7.8. He is an AI Engineer with hands-on experience in full-stack web development, RAG-based Generative AI systems, local LLM deployment, and machine learning applications.
He aims to pursue a professional career in AI Engineering with a focus on practical AI integration, system automation, and LLM-based applications.

Technical Skills:

- Programming Languages: Python, Java, TypeScript
- AI/ML Frameworks: PyTorch, scikit-learn, LangChain, Ollama, ChromaDB
- Web Technologies: MERN Stack (MongoDB, Express.js, React, Node.js)
- Cloud & DevOps: Docker, Google Cloud Platform (GCP), Git/GitHub
- Databases: SQL, SQLite3, MongoDB

Experience:

Freelance AI Developer (Q3Robotics, USA) — *Oct 2025 – Present*

- Developing a restaurant management software for multiple locations.
- Integrating AI-based conversational modules for handling customer orders.

Software Developer — Sedris Vision — *Jul 2024 – Jun 2025*

- Built an accessibility-first retail platform tailored for visually impaired users.
- Worked on improving accessibility interfaces and usability.

Web Developer — Uttam Attires — *Jan 2024 – Jul 2024*

- Developed an online retail store.
- Implemented an AI-powered chatbot for enhanced customer interaction.

Projects:

1. CLI Assistant for Docker
A command-line AI assistant designed to help developers with Docker queries.
Tech: Python, LangChain, Ollama, ChromaDB
Repo: github.com/devesh-saini/Semantic-Search

2. RAG Agent
An AI agent built with retrieval-augmented generation capabilities for contextual responses.
Tech: Python, LangChain, Ollama, ChromaDB
Repo: github.com/devesh-saini/Terminal-Agent

3. Regression Models
Implemented and compared multiple regression algorithms (Linear, Multiple, Polynomial, SVR).
Tech: Python, scikit-learn, NumPy, Pandas, Matplotlib

Research

Paper: “Recent Advancement in Blockchain: A Study”

- Explored upcoming advancements in blockchain, including Block-Mesh, a proposed 2D blockchain model.
- Published on ResearchGate

Leadership & Achievements

- Campus Technology Officer, 10x Club Student Chapter
- Core Technical Member, Google Developers Group (GDG), CGC-COE
- 3rd Position, National Science Day – Project Display Category
- Delivered multiple technical talks on topics such as *AI and the Future*, *Cyber Security Architecture*, and *Virtual & Augmented Reality*.

Certifications and Professional Learning

I’ve completed a diverse set of certifications that reflect my deep interest in Artificial Intelligence, Machine Learning, Programming, and Cybersecurity. My learning journey focuses on bridging theory with practical application — especially in Generative AI, Prompt Engineering, and AI-powered development workflows.

Google Cloud & Gemini AI Certifications:

1. Explore Generative AI with the Gemini API in Vertex AI – I learned to use Google’s Gemini API for building generative AI applications on Vertex AI.
2. Develop GenAI Apps with Gemini and Streamlit – I built and deployed interactive GenAI apps integrating Gemini APIs with Streamlit.

3. Inspect Rich Documents with Gemini Multimodality and Multimodal RAG – I explored how multimodal retrieval-augmented generation works for analyzing both text and visual data.
4. Build Real World AI Applications with Gemini and Imagen – This helped me learn to combine Gemini (text) and Imagen (vision) models for end-to-end AI systems.
5. Prompt Design in Vertex AI – I studied the process of prompt engineering and structuring prompts for consistent outputs within Vertex AI.

Machine Learning and Programming Certifications:
6. Machine Learning with Python (freeCodeCamp) – Covered supervised and unsupervised ML models using scikit-learn and pandas.
7. CS50 Python (Harvard) – Strengthened my understanding of Python fundamentals and problem-solving techniques.
8. CS50x: Introduction to Computer Science (Harvard) – Built a solid foundation in algorithms, data structures, and computer science logic.
9. Foundational C# with Microsoft (freeCodeCamp x Microsoft) – Learned core C# syntax, object-oriented principles, and .NET fundamentals.
10. Java Course: Mastering the Fundamentals (Scaler) – Focused on OOP, control structures, and exception handling in Java.
11. MySQL (Harvard) – Gained hands-on experience in database design, normalization, and SQL queries.
12. CompTIA Security+ (Udemy) – Understood cybersecurity fundamentals like encryption, network defense, and vulnerability analysis.

Event Organization and Technical Leadership

I've always enjoyed community-driven technology initiatives, where I get to collaborate with peers and mentor others. Through Google Developer Group (GDG) activities and technical events, I've honed my leadership and organizational skills while learning to coordinate at scale.

Google Solution Challenge Bootcamp – India Series (CGC Landran, Punjab)

I had the privilege to co-organize the final bootcamp of the India series of the Google Solution Challenge, hosted at Chandigarh Group of Colleges, Landran, under the banner of Google Developer Groups (GDG).
The event was a perfect blend of collaboration, passion, and innovation, where developers, students, and mentors came together to build real-world solutions using Google technologies.

- As part of the organizing team, I was responsible for coordinating logistics, planning sessions, and ensuring smooth execution throughout the event.
- I collaborated with mentors, speakers, and student leaders to curate high-value sessions on ideation, product design, and solution-building strategies.
- The bootcamp featured inspiring talks from industry experts such as Chandra Sekaran, Aashi D., Prasad Seth, Subhrajyoti Sen, Shradha Gurjar, and Uzma Faridi.

    - Uzma Faridi guided participants through ideation, project structuring, and best practices for the Solution Challenge.
    - Shradha Gurjar shared practical insights on using Google technologies to address real-world issues.
- I worked alongside a dedicated team that included Deepti Midha, Kabir Phutela, Harmanpreet Kaur, Manik, Aashi Raghuwanshi, Kanan Kango, Disha Rani, Divyansh Kaushik, Piyush Gupta, Manan Punia, Harshit Singla, Yash Raj Singh, Kushal Bhardwaj, and Abhinav Uniyal, mentored by Dr. Kapil Mehta.

The event concluded the India GDG Bootcamp series and left a lasting impression on everyone involved. For me, it reinforced how collaboration and technology can inspire people to push boundaries and innovate.

Speaker at “Blockchain Breakthrough” Event

I was honored to be invited as a speaker at the *Blockchain Breakthrough* event — an incredible opportunity to engage with an audience of over 70 tech enthusiasts about cybersecurity and digital resilience.

- My session, titled “The Fundamentals of Cybersecurity Architecture and Principles,” focused on simplifying complex security concepts into practical, real-world applications.
- I discussed topics such as secure system design, access control, and the importance of integrating security measures into modern software development.
- Condensing this topic into a 30-minute presentation was a rewarding challenge, and the positive feedback from attendees made the experience even more fulfilling.

Speaking at this event gave me the chance to share knowledge while also learning how to communicate intricate technical ideas in an accessible and engaging way.

Personal Reflection

Through these experiences, I’ve realized that my passion lies in building AI systems, leading developer communities, and sharing knowledge with others. Whether it’s through Generative AI projects, speaking engagements, or organizing large-scale events, I enjoy connecting innovation with real-world impact.

Articles & Blogs — Written by Me (Devesh Saini)

Exploring Generative AI with the Vertex AI Gemini API: My Hands-On Experience

Published: June 3

In this article, I share my hands-on journey of exploring Google Cloud’s Vertex AI Gemini API. I explain how Gemini represents a significant shift in the way developers can build Generative AI applications—covering text generation, multimodality, and real-world integrations.

I discuss setup, workflow, and practical insights gained while experimenting with Gemini models, showing how easily they can be incorporated into everyday development workflows to create intelligent and context-aware systems.

Inspecting Rich Documents with Gemini: My Dive into Multimodality and RAG

Published: June 3

After earning the Google Cloud Skill Badge “Inspect Rich Documents with Gemini Multimodality and Multimodal RAG,” I wrote this post to document my experience.

Here, I dive into **how multimodal retrieval-augmented generation (RAG)** combines images, documents, and text in one workflow using Gemini. I share real examples, configurations, and the benefits of unifying visual and textual intelligence for smarter applications.

Develop GenAI Apps with Gemini and Streamlit

Published: May 11

This article explores how Gemini APIs can be integrated with Streamlit to rapidly prototype and deploy AI-driven applications.

I walk through the process of building a smart assistant UI, connecting it to Gemini for contextual reasoning, and deploying it with minimal effort—bridging Generative AI and modern web frameworks. It’s a practical guide for developers looking to turn LLM ideas into interactive tools.

Building Real-World AI Applications with Gemini and Imagen

Published: April 30

In this blog, I share what I learned while combining Gemini (text) and Imagen (vision) to build end-to-end, real-world AI applications.

I reflect on the process of integrating multimodal APIs, handling data pipelines, and creating applications that understand and generate across media. The post summarizes my takeaways from building something tangible with Google Cloud’s latest AI ecosystem.

Leveling Up with Prompt Design in Vertex AI

Published: April 29

Prompt design is an essential skill when working with LLMs. In this article, I share my personal learning from the Google Cloud course “Prompt Design in Vertex AI.”

I discuss structured prompting strategies, testing methods, and real-world scenarios where prompt tuning dramatically changes model output quality. It’s both a reflection and a resource for developers stepping into LLM-driven app design.

I Secured a Remote Internship: Here’s How It Happened

Published: August 21, 2024

In this personal post, I describe how I landed my first remote internship opportunity. I outline how I prepared my portfolio, positioned my technical skills, reached out to companies, and communicated effectively during the process. The blog serves as honest advice for students and early-career developers aiming to break into tech roles.

Understanding SOLID Principles in Software Development

Published: August 16, 2024

This article dives into the SOLID design principles, explaining how each principle (Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion) contributes to clean, maintainable software.

I present practical code examples and discuss how following SOLID improves scalability and testing in modern applications.

Evolution of Software Architecture

Published: April 4, 2024

In this post, I trace the journey of software architecture—from monoliths to microservices, event-driven design, and the emerging AI-centric architectures.

I emphasize how architecture patterns evolve with hardware, scalability needs, and user expectations, and what that means for future developers.

Next.js Folder Structure: A Developer’s Guide

Published: March 26, 2024

A concise and friendly guide where I explain the Next.js project folder structure, file conventions, and best practices for organizing large-scale React-based projects.

This blog was written to help beginners and intermediate developers understand how structure impacts scalability, maintainability, and developer experience.

Networking Series: OSI Model, UDP Deep Dive, and TCP and Beyond

In this multi-part networking series, I break down core internet protocols and architectures:

OSI Model – A foundational explainer of how data moves through network layers, aimed at simplifying complex networking concepts.

UDP Deep Dive – I unpack how the User Datagram Protocol functions, its trade-offs versus TCP, and how modern protocols like QUIC improve reliability without losing performance.

TCP and Beyond – A detailed look at how TCP connections actually work in practice—covering handshakes, congestion control, and modern evolutions of network reliability.

Summary

Across all my writings, I aim to make complex technical ideas accessible and practically useful for developers. Whether I'm discussing AI, software design, or networking fundamentals, my goal is to share insights from hands-on experience and continuous learning.